

The Namibian cheetah (*Acinonyx jubatus jubatus*), the world's fastest land-dwelling animal, hunts across the African grasslands using high-speed sprints. Cheetahs possess unique anatomical features specialized for speed, including an aerodynamic skull, elongated legs, and an enlarged heart. These physiological adaptations have occurred over time and allow cheetahs to achieve speeds greater than 100 km/h. However, due to their small body and slender limbs, cheetahs (50–64 kg) are limited with regard to the size of the animals they can successfully hunt and therefore typically prey on slightly smaller species that inhabit open grasslands.

It is believed that around 12,000 years ago, Namibian cheetahs experienced an environmental catastrophe that drastically reduced their population. The small number of surviving cheetahs began mating with one another, which led to severe inbreeding depression. Over time, the offspring of subsequent generations exhibited decreased fitness, reduced fecundity (number of offspring), and sustained a loss of major histocompatibility complex (MHC) allele diversity. To study if low MHC diversity correlates with low fitness in cheetahs, scientists analyzed the genetic diversity and disease prevalence in a group of free-ranging and captive male cheetahs.

Experiment 1

MHC allele genotyping and analysis was performed in wild (free-ranging) and captive male cheetahs (Figure 1). Because MHC diversity is strongly correlated with genetic diversity, MHC diversity is used as a marker for an organism's overall genetic variability.

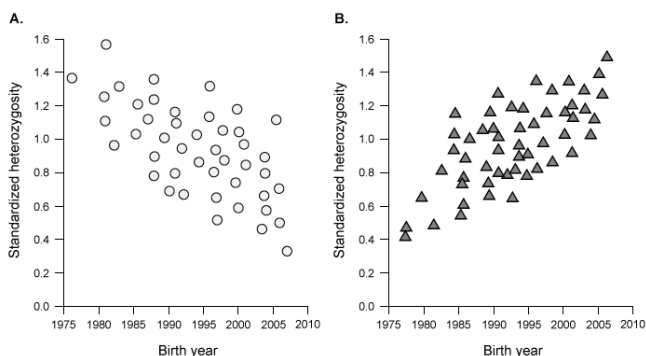


Figure 1 Evaluation of heterozygosity in (A) wild and (B) captive male cheetahs born between 1976 and 2007 (Note: Each data point represents a single male cheetah.)

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Disease prevalence was evaluated in wild and captive male cheetahs by quantitative analysis of symptoms relating gastrointestinal, liver, and kidney disease. The percentages of cheetahs affected by each disease are shown in Figure 2.

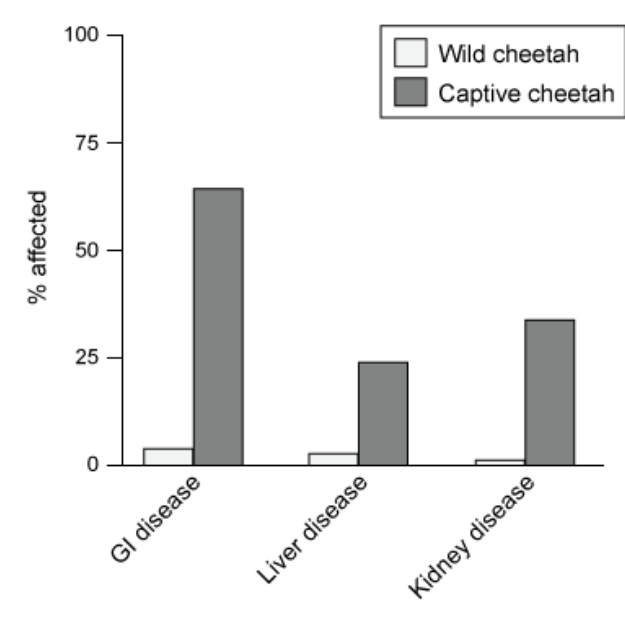


Figure 2 Evaluation of disease prevalence in wild and captive male cheetahs

Experiment 3

Fecal cortisol levels were noninvasively evaluated in wild and captive cheetahs over a period of 6 months (Figure 3).

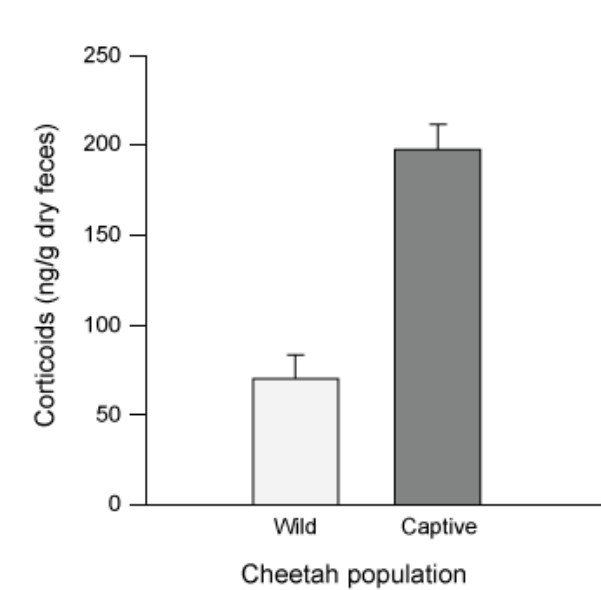


Figure 3 Average fecal cortisol concentrations of wild and captive male cheetahs

The population of cheetahs that survived the environmental catastrophe that

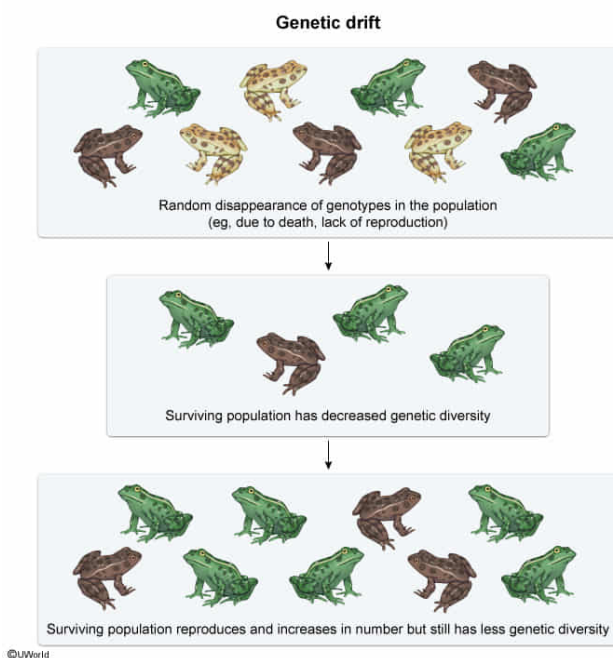
occurred 12,000 years ago were at greater risk of extinction because of:

- ☐ A. changes in allele frequencies due to natural selection. [29%]
- ✓ ☒ B. changes in allele frequencies due to random chance. [46%]
- ☐ C. changes in allele frequencies due to inbound migration from unaffected populations. [7%]
- ☐ D. changes in allele frequencies due to random mating. [16%]

Correct

46% answered correctly

Explanation:



Genetic drift describes the natural **variations** in allele frequencies of a population due to **random** genetic changes that are *not* related to natural selection (eg, sampling error, chance event). Although all populations are affected by genetic drift, its effects are **more significant** in **smaller populations**, which have a **reduced gene pool**. The smaller gene pool cannot buffer random (good or bad) variations in allele frequencies that occur due to chance events. In other words, the probability of a beneficial allele being *removed* by genetic drift *increases* for small populations with low genetic diversity. Similarly, drift **increases the probability** of a **deleterious allele** becoming fixed within the population.

According to the passage, cheetahs experienced a catastrophic event that drastically reduced their population size (ie, a **bottleneck event**). Bottlenecks

(due to environmental events or human action) **greatly reduce the genetic diversity** of a population. Consequently, the smaller population of cheetahs has a **reduced ability to buffer** the **negative** impacts of **random changes** in allele frequencies (genetic drift) that may result in **extinction**.

(Choice A) **Natural selection** is an evolutionary mechanism by which *beneficial* alleles are selected for while unfavorable traits are selected against. These selected changes alter the population's allele frequencies, improving the species' fitness. Consequently, these populations have an *improved* probability for *survival* rather than an increased probability for extinction.

(Choice C) **Gene flow**, or the changes in allele frequency due to inbound migration, *increases* genetic diversity by introducing new alleles to the gene pool. Therefore, small populations with low genetic diversity are more likely to *benefit* from gene flow and therefore are less likely to go extinct.

(Choice D) Random mating increases a population's genetic diversity because of the inherent *variations* that arise during meiosis (recombination) and sexual reproduction (fusion of gametes). Therefore, the variations or changes to the allele frequencies due to random mating will primarily be *beneficial* to a population as a whole.

Educational objective:

Genetic drift is a mechanism of evolution; however, unlike natural selection, the variations in allele frequencies occur randomly by chance. Because smaller populations have a smaller gene pool, the random (good or bad) variations due to genetic drift cannot be buffered.

Time spent:0 sec

Subject:

Biology

QID:401464

System:

Genetics and Evolution

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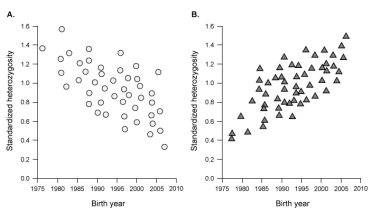


Figure 1 Evaluation of heterozygosity in (A) wild and (B) captive male cheetahs born between 1976 and 2007 (Note: Each data point represents a single male cheetah.)

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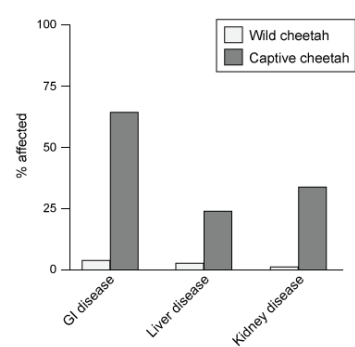


Figure 2 Evaluation of disease prevalence in wild and captive male cheetahs

Experiment 3

Fecal cortisol levels were noninvasively evaluated in wild and captive cheetahs over a period of 6 months (Figure 3).

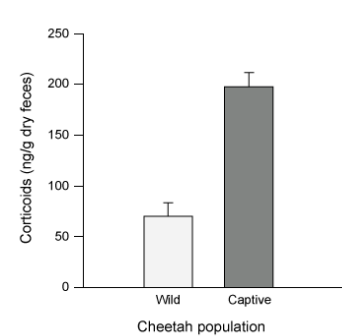


Figure 3 Average fecal cortisol concentrations of wild and captive male cheetahs

Which of the following statements best explains the trend shown in Figure 1? (Note: Captive cheetahs are generally housed individually.)

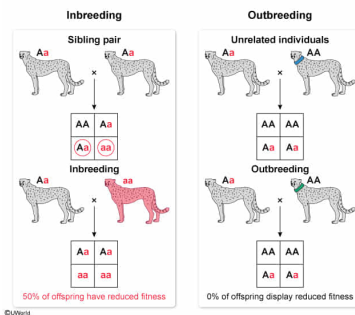
- ☐ A. Captive cheetahs are bred with close relatives. [15%]
- ☒ B. Captive cheetahs are selectively outbred. [63%]
- ☐ C. Captive cheetah populations are in Hardy-Weinberg equilibrium. [8%]

☐ D. Captive cheetahs have increased fecundity. [12%]

Correct

64% answered correctly

Explanation:



The passage states that the event that occurred 12,000 years ago resulted in **inbreeding depression** (decreased heterozygosity and reduced fitness) of Namibian cheetahs. Typically, individuals within a sufficiently large population will **outbreed** (mate with nonrelatives) because of the additional fitness that outbreeding confers. Outbreeding has an evolutionary advantage because the behavior introduces **new genetic material** and reduces the probability of passing familial genetic abnormalities to offspring.

Figure 1 shows that the heterozygosity of wild-born male cheetahs has decreased over time. Decreased heterozygosity is associated with inbreeding; therefore, it can be concluded that free-ranging African cheetahs are inbred. In contrast, the **heterozygosity** of captive-born male cheetahs has **increased**. These results indicate that conservation groups are selectively **outbreeding** captive cheetahs with individuals from distinct populations.

(Choice A) Inbreeding results in decreased heterozygosity. The captive cheetah population has *increased* heterozygosity, *not decreased*; therefore, it can be concluded that the captive cheetahs are bred with individuals that are *not* closely related to each other.

(Choice C) When a population is in **Hardy-Weinberg equilibrium**, allelic frequencies do not change from one generation to another. As such, heterozygosity in Hardy-Weinberg equilibrium is neither decreased nor increased but *remains the same* over time. The captive cheetahs show changing allelic frequencies, so they cannot be in Hardy-Weinberg equilibrium.

(Choice D) Although increased heterozygosity is typically associated with increased fitness and increased fecundity (number of offspring an individual can have), Figures 2 and 3 show that captive cheetahs have increased incidence of diseases and stress hormone cortisol. These data indicate that captive cheetahs have *decreased* fitness and likely will have *decreased fecundity*. Consequently, the increased heterozygosity is not likely explained by an increase in fecundity.

Educational objective:

Inbreeding results in decreased heterozygosity (genetic diversity), reduced fecundity, and reduced fitness. Species that mate with nonrelatives (outbreed) increase their fitness because the introduction of new genetic material results in increased heterozygosity.

Time spent: 0 sec
Subject:
Biology

QID: 401465
System:
Genetics and Evolution

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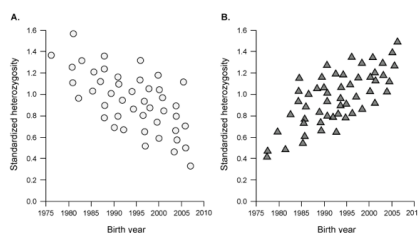


Figure 1 Evaluation of heterozygosity in (A) wild and (B) captive male cheetahs born between 1976 and 2007 (Note: Each data point represents a single male cheetah.)

Experiment 2

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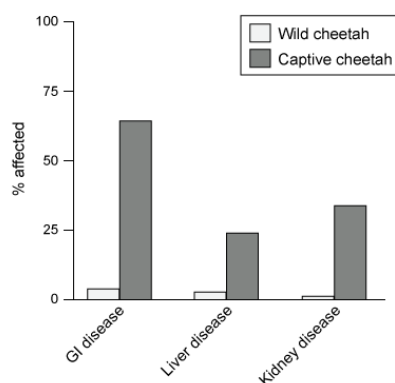


Figure 2 Evaluation of disease prevalence in wild and captive male cheetahs

Experiment 3

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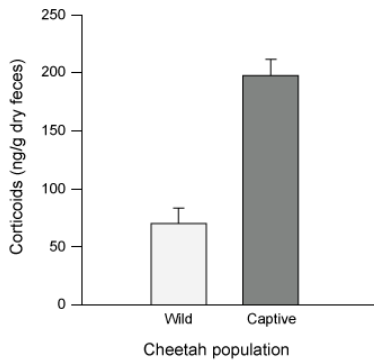


Figure 3 Average fecal cortisol concentrations of wild and captive male cheetahs

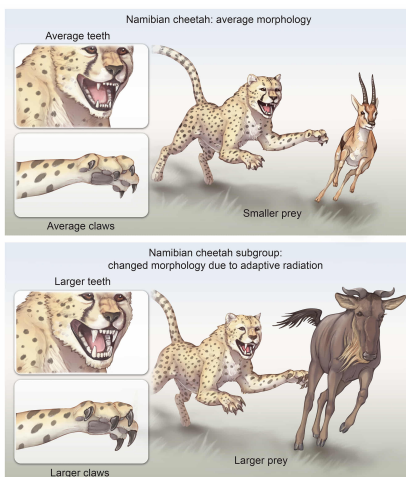
Assume that a subgroup of wild Namibian cheetahs was found to hunt larger prey that weigh more than 100 kg. The individuals that make up this subgroup possess larger claws and teeth compared to average Namibian cheetahs and retain the ability to naturally mate with them. These morphological and behavioral changes reflect:

- ☐ A. the selection of average over extreme phenotypes. [2%]
- ☐ B. how extreme characteristics are favored over average ones. [21%]
- ☒ C. diversification of characteristics to better fill an ecological niche. [73%]
- ☐ D. a new species. [2%]

Correct

74% answered correctly

Explanation:



Adaptive radiation is the process of **diversifying characteristics** in a subgroup of individuals from a **single species**.

Adaptive radiation reduces intraspecific **competition**, the competition for resources by members of a single species. As a result, **fitness is improved** for the entire species because the subgroup has a new role within an ecological community (niche) that is different from the rest of the species. Adaptive radiation can eventually lead to speciation, or the formation of a new species, if the subgroup continues to diverge.

According to the passage, Namibian cheetahs typically hunt species that are slightly smaller than themselves. In this scenario, the subgroup of cheetahs developed larger claws and teeth that enable them to hunt large prey exclusively. Consequently, the subgroup of cheetahs fills a new ecological niche and reduces competition by hunting prey that differs from the average cheetah hunting behavior.

(Choice A) In **stabilizing selection**, phenotypes are narrowed toward an average, homogeneous phenotype by selecting against extreme phenotypes; as a consequence, diversity is decreased within the population. Larger claws and teeth are not average phenotypes; therefore, these traits would not reflect stabilizing selection on the population.

(Choice B) **Disruptive selection** results in the selection of 2 extreme phenotypes that differ from the average. In disruptive selection, the average phenotype is selected against while 2 extreme phenotypes are selected for. Although the subgroup of cheetahs has larger claws and teeth, the average cheetah phenotype is not being selected against; therefore, this

scenario *does not* reflect disruptive selection.

(Choice D) **Speciation** is the evolutionary process of forming a new species from a previously existing species. This process occurs over many generations and results in organisms that *cannot interbreed* with the ancestral species. Although the subgroup of cheetahs developed different characteristics, it retained the ability to mate with Namibian cheetahs and is therefore *not* considered a separate species.

Educational objective:

Adaptive radiation is the process of diversifying characteristics (eg, claw and teeth size) to better fill an ecological niche. Adaptive radiation can eventually lead to speciation if the subgroup continues to diverge and loses the ability to interbreed with individuals from the original species.

Time spent:0 sec

Subject:
Biology

QID:401466

System:
Genetics and Evolution

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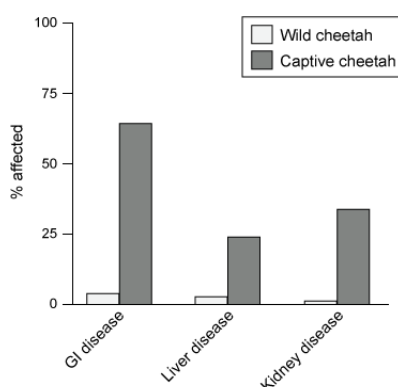


Figure 2 Evaluation of disease prevalence in wild and captive male cheetahs

Experiment 3

Fecal cortisol levels were noninvasively evaluated in wild and captive cheetahs over a period of 6 months (Figure 3).

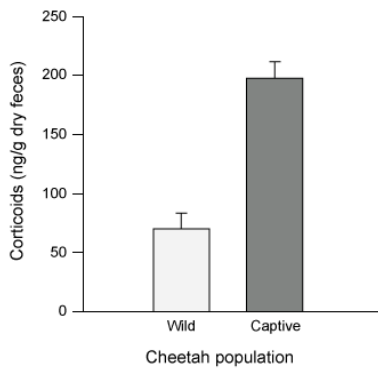


Figure 3 Average fecal cortisol concentrations of wild and captive male cheetahs

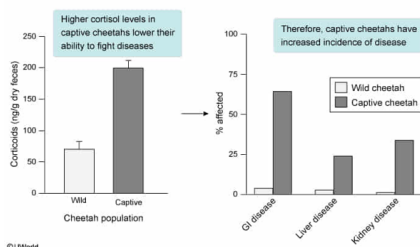
Researchers concluded that captive cheetahs have reduced fitness because of their increased susceptibility to diseases. What can best explain the decreased fitness of captive cheetahs described in the passage?

- ☐ A. Captive cheetahs have increased MHC diversity. [7%]
- ☐ B. Captive cheetahs have decreased MHC diversity. [34%]
- ☒ C. Captive cheetahs experience increased environmental stress. [47%]
- ☐ D. Captive cheetahs experience decreased environmental stress. [10%]

Correct

48% answered correctly

Explanation:



In response to **stress**, the hypothalamus stimulates the pituitary gland to release adrenocorticotrophic hormone (ACTH). As a result of ACTH release, the adrenal cortex secretes **glucocorticoid** hormones such as **cortisol**. Cortisol mediates stress responses by increasing blood glucose levels and decreasing inflammation and protein synthesis. Chronic exposure to cortisol can **negatively impact immunologic responses**, increase the susceptibility for diseases, and affect reproductive health (fecundity).

The data in the passage shows that captive male cheetahs have relatively higher susceptibility to diseases despite having **increased genetic diversity**. As a result, it can be **concluded** that **other factors** besides genetics determine the **fitness** of captive male cheetahs. Indeed, according to Figure 3, captive cheetahs have **increased** fecal **cortisol** levels. Therefore, the **decreased fitness** can be attributed to increased cortisol, which is a **marker** of **increased environmental stress**.

(Choice A) MHC alleles encode surface receptors on all nucleated cells that play a role in the activation of the adaptive immune system. Typically, increased MHC variation would *enhance* an organism's ability to combat a variety of infections and diseases and would normally be associated with *increased* fitness. Therefore, increased MHC variation does *not* explain the *decreased* fitness of captive cheetahs.

(Choice B) Although a decrease in MHC variation and genetic diversity might otherwise explain a decrease in fitness, Figure 1 of the passage shows that captive cheetahs have *increased* MHC variation and genetic diversity.

(Choice D) Cortisol is a biological marker of stress. Because captive cheetahs have increased fecal cortisol concentrations compared to free-ranging cheetahs, it is likely that the environment of captive cheetahs (confinement) induces stress. This heightened environmental stress in captivity leads to an increase in cortisol levels that ultimately reduces immunocompetence and fitness.

Educational objective:

Major histocompatibility complexes (MHCs) are cell surface receptors in vertebrates that present foreign molecules to elicit an immune response. MHC variations are evolutionarily important to an organism's fitness because they influence the organism's ability to combat a variety of infections and diseases. Cortisol, a hormone that negatively impacts immunocompetence, increases in response to stress.

Time spent:0 sec**Subject:**

Biology

QID:401467**System:**

Genetics and Evolution

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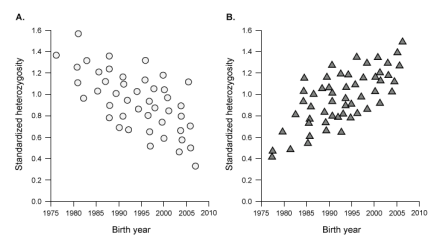


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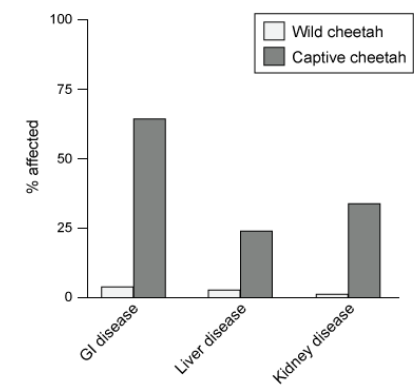


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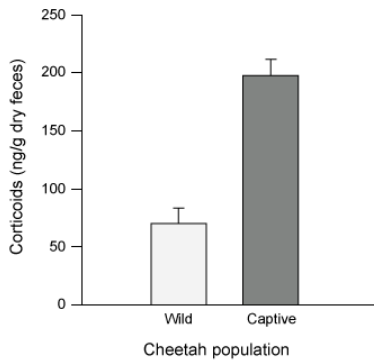


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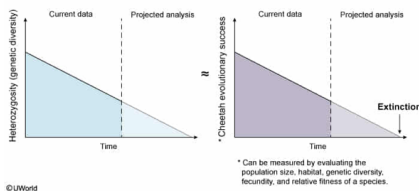
If wild Namibian cheetahs continue breeding as described in the passage, what is most likely to occur to their population size in the long term?

- ☐ A. Increased population size with equal sex ratio [6%]
- ☐ B. A sharp increase followed by a slight decrease in population size [11%]
- ☒ C. Elimination of the wild Namibian cheetah species [78%]
- ☐ D. Stable population size consisting of mostly females [4%]

Correct

78% answered correctly

Explanation:



The passage describes wild cheetahs' breeding patterns as **inbreeding**, the mating of closely related individuals. Inbreeding leads to **inbreeding depression**, which is characterized by loss of genetic diversity (heterozygosity), reduced fecundity (number of offspring), and **decreased fitness**.

Fitness is reduced in **inbreeding populations** because inbred offspring have an increased **probability** of being **homozygous** for **deleterious recessive traits** due to the low genetic diversity between their parents. Individuals with low genetic diversity in MHC alleles have low fitness because it reduces their ability to fight novel diseases and pathogens. Therefore, repeated generational inbreeding within the cheetah population will most likely lead to decreased heterozygosity, reduced reproductive success, and ultimately **extinction** or the **elimination** of the cheetah **species**.

(Choices A and D) Neither sex ratios nor female cheetahs are mentioned in the passage; therefore, no conclusions can be drawn about sex ratios or the number of females within the population due to long-term inbreeding. In addition, the population of free-ranging cheetahs demonstrates decreased heterozygosity, which is indicative of decreased fecundity and fitness. The most likely long-term impact of inbreeding depression is *decreased* (not stabilized or increased) population size.

(Choice B) If wild cheetahs continue the mating pattern described in the passage, the small *inbred* population is unlikely to increase in size over a long period because inbreeding *reduces fitness* and *increases the probability of extinction*. The introduction of new, genetically distinct individuals would be necessary to increase the cheetah's genetic diversity, fitness, and population size.

Educational objective:

Inbreeding depression is characterized by decreased genetic diversity, reduced fecundity, and decreased fitness. As genetic diversity decreases within a population, the frequency of homozygous deleterious recessive genes increases within that population. Consequently, inbred populations have an increased probability of becoming extinct.

Time spent: 0 sec

Subject:
Biology

QID: 401468

System:
Genetics and Evolution